Having trained many scientists who work on LIGO, in 2000 I turned my own research in other directions. But I watch eagerly as LIGO and its international partners near maturity and near their first detections of gravitational waves.

The Warped Side of the Universe

Interstellar is an adventure in which humans encounter black holes, wormholes, singularities, gravitational anomalies, and higher dimensions. All these phenomena are "made from" warped space and time, or are tied intimately to that warping. This is why I like to call them the "warped side of the universe."

We humans, as yet, have very little experimental or observational data from the universe's warped side. That's why gravitational waves are important: they are made from warped space, and so they are the ideal tool for probing the warped side.

Suppose you had only seen the ocean on a very calm day. You would know nothing of the heaving seas and breaking ocean waves that come with a huge storm.

That is similar to our knowledge, today, of warped space and time. We know little about how warped space and warped time behave in a "storm"—when the shape of space is oscillating wildly and the rate of flow of time is oscillating wildly. For me this is a fascinating frontier of knowledge. John Wheeler, the creative coiner we met in earlier chapters, dubbed this "geometrodynamics": the wildly dynamical behavior of the geometry of space and time.

In the early 1960s, when I was Wheeler's student, he exhorted me and others to explore geometrodynamics in our research. We tried, and failed miserably. We didn't know how to solve Einstein's equations well